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Chapter 9

Dynamic Routing Protocols

In the previous chapter, you were exposed to static routing, which involved manually adding routes to the router's routing table. This can be a long, involved, tedious task when working with large networks. That said, dynamic routing protocols can be loaded on the router so that the routers on the network exchange routing table information in order to build up the routing table. This chapter presents practice questions that test your knowledge of two routing protocols you must be familiar with for the ICND1 exam — RIP and OSPF.

The Problems You'll Work On

In this chapter, you'll review questions concerning the following topics:

- Understanding dynamic routing protocols
- Configuring RIP and RIPv2
- Grasping the basics of configuring OSPF

What to Watch Out For

Don't let common mistakes trip you up; watch for the following when working with these questions:

- To display your routing table, use the `show ip route` command. To view routing protocol details, you use the `show ip protocols` command.
- Distance vector routing protocols send the entire routing table in an update at regular intervals to neighboring routers, while link state routing protocols are aware of the entire network topology and the state of the links. Distance vector routing protocols use the hop count metric, while link state uses the availability of the link and bandwidth.
- RIP is an example of a distance vector routing protocol. To enable RIPv1, you use the `router rip` command to navigate to the router prompt and then enable RIP on each network interface with the network command. If you wish to use RIPv2, you then type the `version 2` command at the router prompt.
- When troubleshooting RIP, you can turn on RIP debugging with the `debug ip rip` command. Be sure to turn off all debugging, when you finish troubleshooting, with the `no debug all` command.

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Introduction to Dynamic Routing Protocols

314–329 Choose the best answer(s).

314. When a router receives a routing table update for a route it does not have in its routing table, what does it do?

(A) Adds the route to the MAC address table

(B) Adds the route to the ARP cache

(C) Adds the route to the routing table

(D) Adds the route as the GWLR

315. Which of the following protocols are distance vector routing protocols supported by multiple vendors and not just Cisco?

(A) OSPF

(B) RIP

(C) IGRP

(D) EIGRP

316. Which of the following protocols are link state routing protocols? (Select all that apply.)

(A) EIGRP

(B) RIP

(C) IGRP

(D) OSPF

317. Which routing protocol is limited to 15 hops?

(A) OSPF

(B) IGRP

(C) RIP

(D) EIGRP

318. A _____routing protocol is a routing protocol that knows only about how many hops away a network is

(A) Distance vector

(B) Layer 2

(C) Link state

(D) Layer 4

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319. Dynamic routing protocols run at what layer of the OSI model?

(A) Layer 1

(B) Layer 2

(C) Layer 3

(D) Layer 4

320. Which of the following is true of distance vector routing protocols? (Select two.)

(A) Routers share routing table with all other routers on the network

(B) Routers share routing table with neighboring routers

(C) Updates are only sent every 60 seconds

(D) Maintains multiple tables in memory — one for neighboring routers, one to store the entire topology, and final table is the routing table.

(E) Sends entire routing table as an update

321. Which of the following are true statements about RIPv2? (Select three.)

(A) It is a classless routing protocol

(B) Maximum hop count of 15

(C) Sends updates with broadcast

(D) RIPv2 supports authentication

(E) Has a high administrative distance than RIPv1

322. (True/False): RIPv1 is an example of a classless routing protocol.

(A) True

(B) False

323. How frequently does RIPv2 send out routing updates?

(A) Every 15 seconds

(B) Every 180 seconds

(C) Every 30 seconds

(D) Every 60 seconds

324. Which of the following is true of a distance vector routing protocol? (Select two.)

(A) Uses a topology table to determine whom to send updates to

(B) Sends periodic updates

(C) Updates routing table based on updates received from neighboring routers

(D) Sends entire routing table to all routers on the network

(E) Uses the shortest path first algorithm to determine best route

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325. OSPF is an example of a _____ routing protocol.

(A) Layer 2

(B) Distance vector

(C) Link state

(D) Layer 4

326. Which of the following is true regarding link state routing protocols? (Select two.)

(A) Routers share routing table information with all other routers on the network

(B) Routers share routing table information with neighboring routers

(C) Updates are only sent every 60 seconds

(D) Maintains multiple tables in memory — one for neighboring routers, one to store the entire topology, and final table is the routing table.

(E) Sends entire routing table as an update

327. Your router receives an update for a route from both RIP and OSP (F) Which route will the router add to its routing table?

(A) RIP

(B) OSPF

(C) Neither

(D) Both

328. What does the term "routing by rumor" refer to?

(A) Broadcasting all routes to neighboring routers

(B) Broadcasting all routes to all routers

(C) Broadcasting static routes to neighboring routers

(D) Sharing routes learned from one neighbor to other neighboring routers

329. Which of the following represents a downfall of distance vector routing protocols over link state?

(A) Convergence time

(B) Sends updates to all routers

(C) Updates are triggered based off events

(D) Updates are delivered using unicast communication

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Configuring RIP and RIPv2

330–341 Choose the best answer(s).

330. Your manager is thinking about upgrading from RIPv1 to RIPv2 and asks about the benefit of RIPv2. What would you say?

(A) Supports a maximum of 30 hops

(B) Sends updates every 90 seconds

(C) Is classful

(D) Supports VLSM

331. Which routing protocol sends the entire routing table to neighboring routers every 30 seconds?

(A) OSPF

(B) RIP

(C) IGRP

(D) EIGRP

332. You wish to see how RIP has been configured on the router. What command would you use?

(A) RouterA#`show ip config`

(B) RouterA#`show protocols`

(C) RouterA#`show rip protocols`

(D) RouterA#`show ip protocols`

333. When looking at the routing table, how do you know which entries have been learned through RIP?

(A) The entry with code R

(B) The entry with code C

(C) The entry with code RIP

(D) The entry with code 1

334. What version of RIP supports VLSM?

(A) RIPv1

(B) RIPv2

(C) RIPv8

(D) RIPv9

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335. What command enables RIP debugging?

(A) `rip debug`

(B) `debug rip`

(C) `debug ip rip`

(D) `enable debug`

336. You need to configure RIPv2 on your router. Which of the following commands would do this?

(A) Use the following commands:

R1>`router rip`

R1>`network 25.0.0.0`

R1>`network 26.0.0.0`

R1>`version 2`

(B) Use the following commands:

R1(config)#`router rip`

R1(config-router)#`network 25.0.0.0`

R1(config-router)#`network 26.0.0.0`

R1(config-router)#`version 2`

(C) Use the following commands:

R1(config)#`router rip 2`

R1(config-router)#`network 25.0.0.0`

R1(config-router)#`network 26.0.0.0`

(D) Use the following commands:

R1(config)#`router rip version 2`

R1(config-router)#`network 25.0.0.0`

R1(config-router)#`network 26.0.0.0`

337. What is the default administrative distance for RIP?

(A) 90

(B) 100

(C) 120

(D) 150

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338. You have three interfaces on the router: one configured for the 27.0.0.0/8, while the other two interfaces are configured with 29.1.0.0/16 and 29.2.0.0/16. If you want to use RIPv2, which of the following represents the least number of commands you would need to use?

(A) Use the following commands:

```
RouterA>enable
RouterA#config term
RouterA(config)#enable rip
RouterA(config)#network 27.0.0.0
RouterA(config)#network 29.0.0.0
RouterA(config)#version 2
```

(B) Use the following commands:

```
RouterA>enable
RouterA#config term
RouterA(config)#router rip
RouterA(config-router)#network 27.0.0.0
RouterA(config-router)#network 29.1.0.0
RouterA(config-router)#network 29.2.0.0
```

(C) Use the following commands:

```
RouterA>enable
RouterA#config term
RouterA(config)#router rip
RouterA(config-router)#network 27.0.0.0
RouterA(config-router)#network 29.1.0.0
RouterA(config-router)#network 29.2.0.0
RouterA(config-router)#version 2
```

(D) Use the following commands:

```
RouterA>enable
RouterA#config term
RouterA(config)#router rip
RouterA(config-router)#network 27.0.0.0
RouterA(config-router)#network 29.0.0.0
RouterA(config-router)#version 2
```

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339. Which of the following is true about RIPng?

(A) You need to enable RIPng on each interface.

(B) RIPng uses broadcast traffic to share routing information.

(C) RIPng is classful.

(D) RIPng supports up to 30 hops.

340. You want to disable sending RIP messages on interface serial 0/1. Which of the following commands would you use?

(A) RouterA(config)#`passive-interface serial 0/1`

(B) RouterA(config-router)#`passive-interface serial 0/1`

(C) RouterA#`passive-interface serial 0/1`

(D) RouterA>`passive-interface serial 0/1`

341. When RIP has multiple routes to a destination with the same hop count, what will it do?

(A) RIP uses the one with the administrative distance of 120.

(B) RIP will load-balance the links to that destination.

(C) RIP uses the second route in the table.

(D) RIP uses the first route in the table.

Basics of Configuring OSPF

342–351 Choose the best answer(s).

342. What is the administrative distance of OSPF?

(A) 110

(B) 90

(C) 120

(D) 115

343. Which of the following would be used as the OSPF router ID if all were configured?

(A) Highest IP address assigned to a loopback interface

(B) Highest IP address assigned to a physical interface

(C) Lowest IP address assigned to a loopback interface

(D) Lowest IP address assigned to a physical interface

344. What type of OSPF router connects one or more areas to the backbone network?

(A) BR

(B) IR

(C) ASBR

(D) ABR

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345. You are troubleshooting OSPF configuration. What command would you use to verify your configuration?

(A) `show ospf`

(B) `show ip ospf`

(C) `show interfaces`

(D) `show controllers`

346. When configuring OSPF on your router, you have specified an interface to be part of area 0. What is the area known as?

(A) Backbone

(B) Branch network

(C) Exterior network

(D) Internet

347. You need to configure OSPF on your router. Which of the following most accurately depicts the commands used?

(A) Use the following commands:

```
router ospf 1
network 192.168.1.0 255.255.255.0 area 0
```

(B) Use the following commands:

```
router ospf 1
network 192.168.1.0 0.0.0.255 area 0
```

(C) Use the following commands:

```
router ospf
network 192.168.1.0 0.0.0.255 area 0
```

(D) Use the following commands:

```
router ospf
network 192.168.1.0 0.0.0.255
```

348. You want to configure your router for OSPF and run it on interfaces that are on the 192.168.1.0 network. Which of the following commands would you use?

(A) Use the following:

```
router ospf 1
network 192.168.1.0 0.0.0.255 area 0
```

(B) Use the following:

```
router ospf 1
network 192.168.1.0 255.0.0.0 area 0
```

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(C) Use the following:

```
router ospf
network 192.168.1.0 0.0.0.255 area 0
```

(D) Use the following:

```
router ospf 1
network 192.168.1.0 255.255.255.255 area 0
```

349. When troubleshooting OSPF, which of the following commands would you use? (Select two.)

(A) `show cdp neighbors`

(B) `show ip protocols`

(C) `show mac-address-table`

(D) `show`

(E) `show ip ospf neighbor`

350. Which of the following commands enables an OSPFv3 process?

(A) `router ospf 5`

(B) `ospf 5`

(C) `ipv6 ospf 5`

(D) `ipv6 router ospf 5`

351. Which of the following commands would you use to change the router ID of an OSPF router?

(A) Use the following:

```
Router rip
Router-id 8.8.8.8
```

(B) Use the following:

```
Router config
Router-id 8.8.8.8
```

(C) Use the following:

```
Router ospf 1
Set Router id 8.8.8.8
```

(D) Use the following:

```
Router ospf 1
Router-id 8.8.8.8
```